M3: semantic item codes and matched permutation controls

Amazon Beauty 2014 · metadata-only tokenization → next-item prediction → frozen test evaluation

A METADATA-ONLY TOKENIZATION

Product metadata

Title · brand
Categories · description

Frozen MiniLM

384-d normalized vectors
Training-catalog items only

Tokenizer fitting objective: metadata structure

RQ reconstructs frozen metadata vectors.
No recommendation-validation or test labels fit tokenizers.
Tokenizer fitting is separate from next-item training.

Fixed hierarchical k-means

3 levels · up to 16 branches
Deterministic within-prefix suffix

Learned residual quantizer (RQ)

3 × 16 codebooks · 32-d latent
Reconstruction + codebook + commitment
90/10 metadata epoch selection
Fresh full-catalog refit

Fit all three RQ tokenizers before recommendation training.

HKM

Semantic item-to-code assignment

HKM-permuted

Whole codeword permutation

RQ

Semantic item-to-code assignment

RQ-permuted

Whole codeword permutation

Freeze all code mappings

4 arms per seed
3 tokens + unique suffix
Common vocabulary
Matched model capacity

frozen item-to-code tables

B NEXT-ITEM TRAINING AND VALIDATION SELECTION

Chronological histories

Last 20 warm items → code tokens
Predict the next reviewed item

4 arms × 3 paired seeds = 12 runs

Same encoder–decoder Transformer
Fresh weights; common vocabulary
Teacher-forced token cross entropy

Validation checkpoint selection

Fixed 1,000 users; maximize Recall@100
MAP@20 ≥ validation popularity
Save fallback if no epoch passes

Freeze all 12 selected checkpoints

Record hashes before any test retrieval
No test-driven epoch or setting changes

Next-item objective: token cross entropy. MiniLM, tokenizers and item-code mappings stay frozen.

selected weights frozen for inference

C FROZEN RETRIEVAL AND HELD-OUT COMPARISONS

Catalog-trie beam search

Beam width 200 → top-200 list
Exclude ALL earlier warm seen items
Approximate autoregressive retrieval

Full eligible warm-target test cohort

Same contexts across arms and seeds
Recall@100 · MAP@20
One saved list → all planned cutoffs

Planned paired contrasts

Primary (Recall@100): HKM − HKM-permuted; RQ − RQ-permuted
Secondary: RQ − HKM
Per-seed paired user bootstrap; pipeline mean ± sample SD

Retain finite poor outcomes and failed guards; do not pool users across seeds as independent observations.

Solid arrows: data/model flow. Dashed arrows: frozen artifacts.

Protocol schematic · no performance results